

#This code will be used to demonstrate proposition 6.16. We demonstrate that the component nj and nk are positive along the curve γ_l

Define nj

$$nj := (x^{-3} - w^{-3}) \cdot w + (x^{-3} - z^{-3}) \cdot z;$$

Define nk

$$nk := (w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3})$$

$$nj := \left(\frac{1}{x^3} - \frac{1}{w^3} \right) w + \left(\frac{1}{x^3} - \frac{1}{z^3} \right) z$$

$$nk := \left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4.

$$parametrization := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, nj))));

simplify(numer(expand(subs(parametrization, nk))));

$$\begin{aligned} & - (b^2 + 1) a (a^2 + 1) (a b + 1) b (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + \\ & \quad - 5 b^7 - 6 b^5 + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 \\ & \quad - 3 b) a^4 + (-2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a \\ & \quad - b^5) \end{aligned}$$

$$\begin{aligned} & 12 a^3 \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \\ & \quad + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\ & \quad - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\ & \quad \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) b^3 \quad (2) \end{aligned}$$

Define $\tilde{n}j$

$$\tilde{n}j := (a^9 b^4 + (-3 b^5 + 2 b^3) a^8 + (3 b^6 - 4 b^4 + b^2) a^7 + (-5 b^7 - 6 b^5 + b^3 + 2 b) a^6 + (3 b^8 + 2 b^6 - 20 b^4 - 4 b^2 + 1) a^5 + (-b^9 + 12 b^5 - 6 b^3 - 3 b) a^4 + (-2 b^8 - 3 b^6 + 2 b^4 + 3 b^2) a^3 + (-3 b^7 - 5 b^3) a^2 + (-2 b^6 + 3 b^4) a - b^5) :$$

Define ñk

$$\begin{aligned} \tilde{n}k := & \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) : \end{aligned}$$

#Define r1

$$\begin{aligned} r1 := & \left(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} \right. \\ & + 6 b^{12}) a^{28} + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} \\ & + 20 b^{28} - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} \\ & - 317080 b^{14} - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (\\ & - 1016 b^{26} - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} \\ & - 216 b^{10} + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} \\ & - 1912 b^{14} + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\ & + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\ & - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\ & - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\ & + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\ & + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \end{aligned}$$

$$\begin{aligned}
& + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\
& + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\
& \left. + 2369 b^6 - 699 b^4 + b^2 + 1 \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\
& \left. - 528 b^4 + \frac{4168}{9} b^2 + 1 \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\
& \left. - 1 \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\
& - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \\
& \left. + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big) :
\end{aligned}$$

#Plot the curves $r_1=0$, $\tilde{n}_j=0$ and $\tilde{n}_k=0$ in region A1.

$r := a - b :$

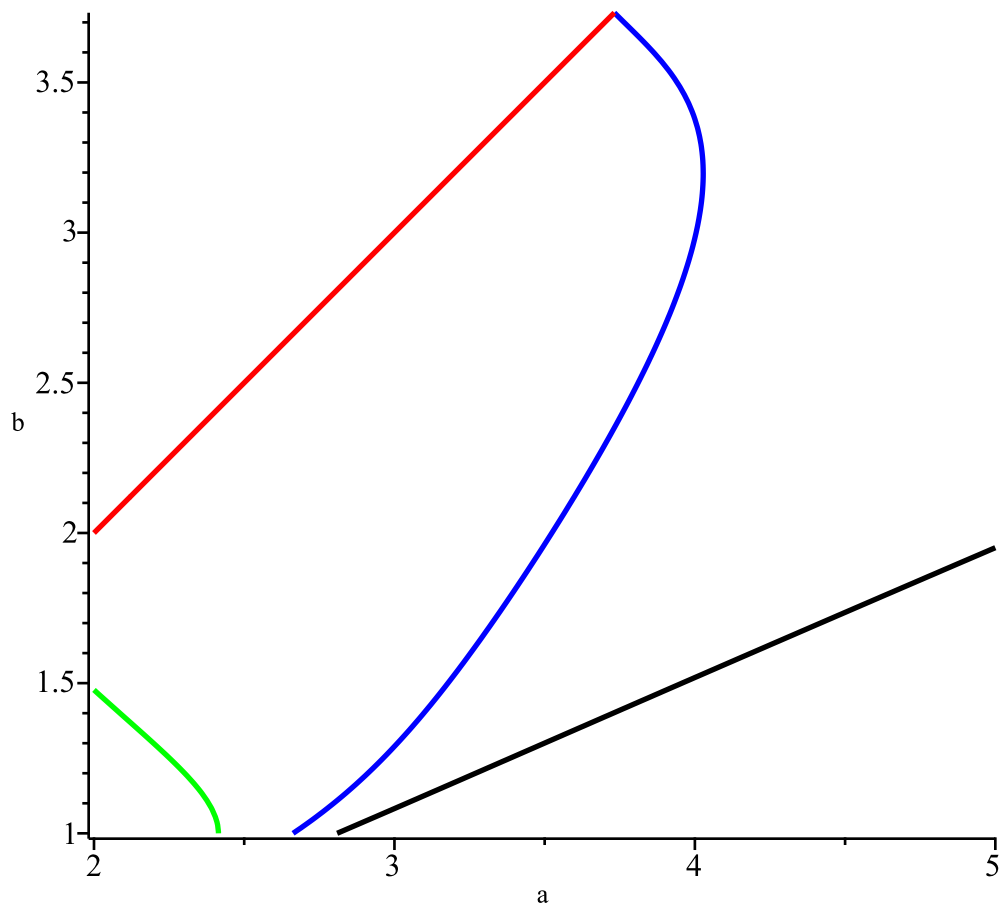
with(plots) :

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implicitplot([r1=0, ñj=0, ñk=0, r=0], a=2..5, b=1..(2+sqrt(3)),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, black, green, red],
  labels=["a", "b"],
  title="Graphic for r1=0, ñj=0 and ñk=0 in A1");

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Graphic for $r_1=0, \tilde{n}_j=0$ and $\tilde{n}_k=0$ in A1



#Compute the resultant between $r1$ and $\tilde{n}j$ with respect to a

$Ra := \text{resultant}(r1, \tilde{n}j, a) :$

#Compute the factors of the resultant Ra and count how many there are

$fator := [\text{op}(\text{factor}(Ra))] :$
 $\text{nops}(bfator);$

8 (3)

#We list below the factors of the resultant Ra . When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, 2 + \sqrt{3})$ where γl is defined

$fator[1];$

65536 (4)

$fator[2];$

b^{144} (5)

$fator[3];$

$\text{sturm}(\text{sturmseq}(fator[3], b), b, 1, (2 + \sqrt{3})); \# \text{number of roots in } (1, (2 + \sqrt{3}))]$
 $9 b^{16} + 192 b^{14} + 1108 b^{12} + 2800 b^{10} + 44686 b^8 - 13536 b^6 + 9540 b^4 + 432 b^2 + 81$
0 (6)

$fator[4];$

$\text{sturm}(\text{sturmseq}(fator[4], b), b, 1, (2 + \sqrt{3})); \# \text{number of roots in } (1, (2 + \sqrt{3}))]$
 $21609 b^{32} + 419832 b^{30} + 4619328 b^{28} + 35381304 b^{26} + 271691620 b^{24} + 2161519832 b^{22}$
 $+ 14661620592 b^{20} + 41509040792 b^{18} + 58790686846 b^{16} + 31859784168 b^{14}$
 $+ 6399720480 b^{12} + 245450664 b^{10} - 21818700 b^8 + 900936 b^6 + 1023408 b^4 + 104328 b^2$
 $+ 3969$
0 (7)

$fator[5];$

$(3 b^2 - 1)^2$ (8)

$fator[6];$

$\text{sturm}(\text{sturmseq}(fator[6], b), b, 1, (2 + \sqrt{3})); \# \text{number of roots in } (1, (2 + \sqrt{3}))]$
 $(3 b^4 - 6 b^2 + 7)^2$
0 (9)

$fator[7];$

$\text{sturm}(\text{sturmseq}(fator[7], b), b, 1, (2 + \sqrt{3})); \# \text{number of roots in } (1, (2 + \sqrt{3}))]$
 $(374017 b^{48} - 1082256 b^{46} - 2821284 b^{44} + 38273024 b^{42} - 137706774 b^{40} - 25011408 b^{38}$
 $+ 2222925500 b^{36} - 11137697376 b^{34} + 34666181511 b^{32} - 79729549344 b^{30}$
 $+ 140202896376 b^{28} - 190367984256 b^{26} + 204746625068 b^{24} - 177535429920 b^{22}$

$$\begin{aligned}
&+ 126486144024\, b^{20} - 74566977600\, b^{18} + 36572797215\, b^{16} - 14813913168\, b^{14} \\
&+ 4932365356\, b^{12} - 1327146624\, b^{10} + 292102794\, b^8 - 53430800\, b^6 + 8765868\, b^4 \\
&- 983136\, b^2 + 45241)^2 \\
&\hspace{15em} 0 \hspace{10em} \textbf{(10)}
\end{aligned}$$

$$\begin{aligned}
&fator[8]; \\
&\hspace{15em} (b^2 + 1)^{66} \hspace{10em} \textbf{(11)}
\end{aligned}$$

#The curves $r1=0$ and $\tilde{n}j=0$ do not intersect.

#Check the sign of $\tilde{n}_j$ at the point $(\gamma_l(1), 1)$

#Locate a range for $\gamma_l(1)$

sturm(sturmseq(eval(r_1 , $b = 1$), a), $a, 2, 2.7$); #number of roots in $[2, 3)$

1

(12)

#Verify that in this interval $\tilde{n}_j(a, 1)$ has a constant sign.

sturm(sturmseq(eval($\tilde{n}_j$, $b = 1$), a), $a, 2, 2.7$); #number of roots in $[2, 3)$

0

(13)

#Calculate the sign of $\tilde{n}_j(\gamma_l(1), 1)$ using the sign of $\tilde{n}_j(2, 1)$.

eval(eval($\tilde{n}_j$, $b = 1$), $a = 2$)

-831

(14)

#The sign of n_j is opposite to that of $\tilde{n}_j$, it follows that $n_j > 0$ along γ_l

[illegible]

[illegible]

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